



RV UNIVERSITY			
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING			
MINI PROJECT SYNOPSIS			
PROGRAM			
USN	1RVU23CSE339	NAME	Prajna
USN	1RVU23CSE292	NAME	Sanjana
USN	1RVU23CSE317	NAME	Nithish Gowda H N
USN	1RVU23CSE282	NAME	Mohan Chandra S S
PROJECT TITLE	LegalTech		
GUIDE NAME	Dr Nagaraja		
DESIGNATION	Professor		

PROBLEM DEFINITION

Legal contracts are often lengthy, complex, and written in highly technical language, making manual review difficult and time-consuming. Critical clauses related to payment terms, liability, termination conditions, and compliance are frequently overlooked, exposing individuals and organizations to significant legal and financial risks.

Traditional contract analysis depends heavily on legal experts and lacks intelligent automation for identifying potential risks. Additionally, conventional document storage mechanisms do not fully guarantee confidentiality, integrity, and trust, leaving sensitive legal documents vulnerable to tampering.

There is a growing need for a secure and intelligent system that can automatically analyze contracts, highlight risks, support negotiation workflows, and permanently preserve finalized agreements in a tamper-proof manner.

OBJECTIVE (s)

1. To develop an AI-based system that automatically extracts and classifies legal contract clauses.
2. To implement a risk scoring engine that assigns risk levels with explanations.
3. To provide an interactive dashboard that visually highlights risky sections for better decision-making.
4. To integrate blockchain technology for immutable storage of finalized contract records.
5. To enable secure digital signing with cryptographic verification for non-repudiation.
6. To support contract versioning and amendments while preserving historical integrity.

METHODOLOGY

The proposed system follows a structured workflow combining artificial intelligence with blockchain technology for secure and intelligent contract management.

Initially, the contract owner uploads a legal document in formats such as PDF, DOCX, or plain text. The system performs text extraction, clause segmentation, classification, and sentence-level risk scoring using an AI engine deployed on an application server.

A risk visualization dashboard highlights clauses using color indicators (high, medium, low risk) and provides AI-generated explanations to help the reviewing party clearly understand potential issues.

During the negotiation phase, the reviewer may request modifications. Each revision generates a temporary cryptographic hash for version tracking while keeping data off-chain. The contract is repeatedly analyzed until both parties reach agreement.

Once accepted, editing is permanently locked, and both parties digitally sign the contract using private keys. A permanent hash of the contract and AI risk report is then recorded on the blockchain along with signatures, timestamps, and version metadata.

This ensures identity verification, data integrity, and non-repudiation while preventing unauthorized modifications. Future verification is performed by comparing the contract hash with the blockchain record to detect any tampering.

The system also supports amendments by archiving the original contract and linking new versions cryptographically, thereby maintaining a complete immutable history.

INNOVATION / CONTRIBUTION TO THE FIELD

The primary innovation of this project is the development of an intelligent, trust-centric contract management ecosystem that merges automated legal reasoning with decentralized verification.

Unlike conventional systems that rely heavily on manual expertise and centralized storage, the proposed solution introduces explainable AI for risk-aware contract interpretation alongside blockchain for immutable record creation. This dual-layer architecture not only strengthens security and transparency but also enables verifiable negotiations, authenticated approvals, and cryptographically linked version control significantly advancing the reliability and scalability of digital legal infrastructure.

Key contributions of the project include:

- Automated lawyer-like contract review that reduces time, cost, and human error
- Clause-level risk detection with explainable AI insights
- Blockchain-based immutability ensuring contracts cannot be altered after acceptance
- Secure digital signature workflow for authentication and non-repudiation
- Cryptographically linked version control for amendments
- A trustless environment that minimizes disputes and potential litigation

By transforming unstructured legal documents into structured, risk-aware insights, the system enables data-driven decision-making and modernizes contract management practices..

HARDWARE / SOFTWARE REQUIREMENTS

Hardware Requirements:

- Personal computer with multi-core processor
- Minimum 8 GB RAM
- Stable internet connectivity

Software Requirements:

- Backend: Node.js + Express
- AI Engine: Python + FastAPI
- Database: Supabase (PostgreSQL)
- Blockchain: Solidity + MetaMask + Ethers.js
- Frontend: Next.js + Tailwind CSS

Signature of the Review Committee
Committee members (Name and Signature)

Signature of the Guide

Signature of Program Director

Signature of the Dean